

Table of Contents

1. Technology Choices:2

2. Column metadata for each table:.....8

3. Cardinality:.....12

4. Information Schema:.....13

 The total size of the database schema:.....13

 The list of tables in the database schema, the number of entries(row count) on each table in this
 schema and size of each table on schema on descending order.....14

 List of foreign key constraints name and unique constraints name on the database schema....15

Technology Choices:

MySQL Community Server 9.2.0 Innovation

MySQL Shell 9.2.0 Innovation

MySQL Workbench 8.0.41

Column metadata for each table:

Table name	Column	Data type/ Max length
User	id(PK)	int auto_increment
	name	varchar(255) not null
	email	varchar(255) unique not null
	phone	varchar(15)
	user_type	enum not null
Host	id(PK)	int auto_increment
	user_id(FK)	int not null
	Bank_account_details	varchar(255) not null
Guest	id(PK)	int auto_increment
	user_id(FK)	int not null
	Payment_method	varchar(255) not null

Table name	Column	Data type/Max length
Location	id(PK)	int auto_increment
	country	varchar(255) not null
	city	varchar(255) not null
	street	varchar(255) not null
	post_code	varchar(255) not null
Property	id(PK)	int auto_increment
	host_id(FK)	int not null
	name	varchar(255) not null
	description	text
	location_id(FK)	int not null
	price_per_night	decimal(10, 2) not null
Payment	id(PK)	int auto_increment
	booking_id(FK)	int not null
	amount	decimal(10, 2) not null
	payment_date	date not null
	payment_method	varchar(255) not null

Table name	Column	Data type/Max length
Booking	id(PK)	int auto_increment
	property_id(FK)	int not null
	guest_id(FK)	int not null
	check_in_date	date not null
	check_out_date	date not null
	total_price	decimal(10, 2) not null
	status	enum not null
Review	id(PK)	int auto_increment
	booking_id(FK)	int not null
	reviewer_id(FK)	int not null
	reviewee_id(FK)	int not null
	rating	int check
	comment	text
	review_date	date not null
Amenity	id(PK)	int auto_increment
	name	varchar(255) not null

Table name	Column	Data type/ Max length
PropertyAmenity	property_id(PK), (FK)	int auto_increment, int not null
	amentiy_id(PK), (FK)	int auto_increment, int not null
Availability	id(PK)	int auto_increment
	property_id(FK)	int not null
	date	date not null
	is_available	boolean not null
Commission	id(PK)	int auto_increment
	booking_id(FK)	int not null
	commission_rate	decimal(10, 2) not null
	commission_amount	decimal(10, 2) not null
Payout	id(PK)	int auto_increment
	host_id(FK)	int not null
	amount	decimal(10, 2) not null
	payout_date	date not null
	approved_by(FK)	int

Table name	Column	Data type/Max length
Message	id(PK)	int auto_increment
	sender_id(FK)	int not null
	receiver_id(FK)	int not null
	message_text	text not null
	timestamp	datetime not null
SocialMedia	id(PK)	int auto_increment
	user_id(FK)	int not null
	platform_name	varchar(255) not null
	profile_link	varchar(255) not null
Dispute	id(PK)	int auto_increment
	booking_id(FK)	int not null
	complaint_text	text not null
	resolved_by(FK)	int
	resolution_status	enum not null

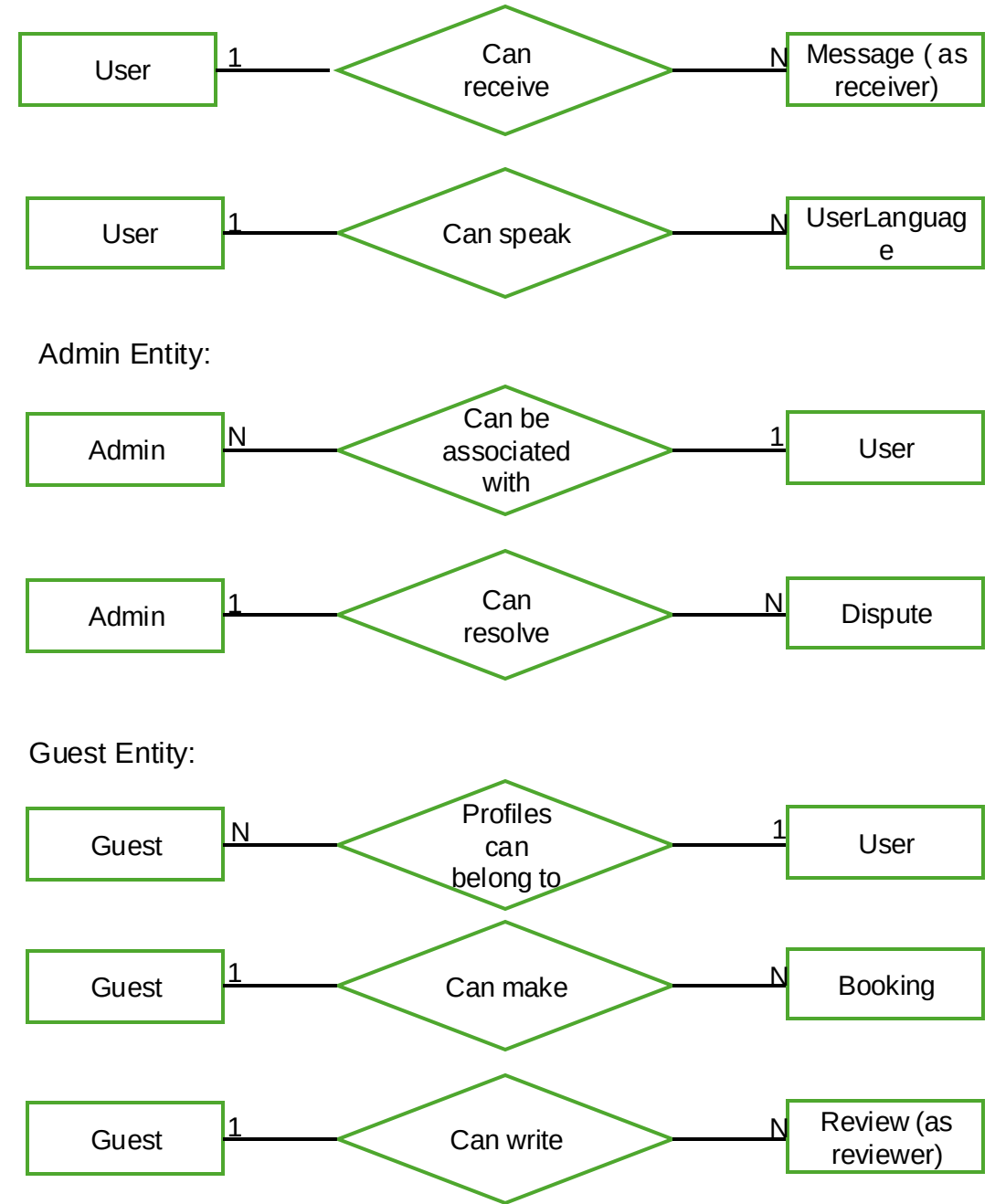
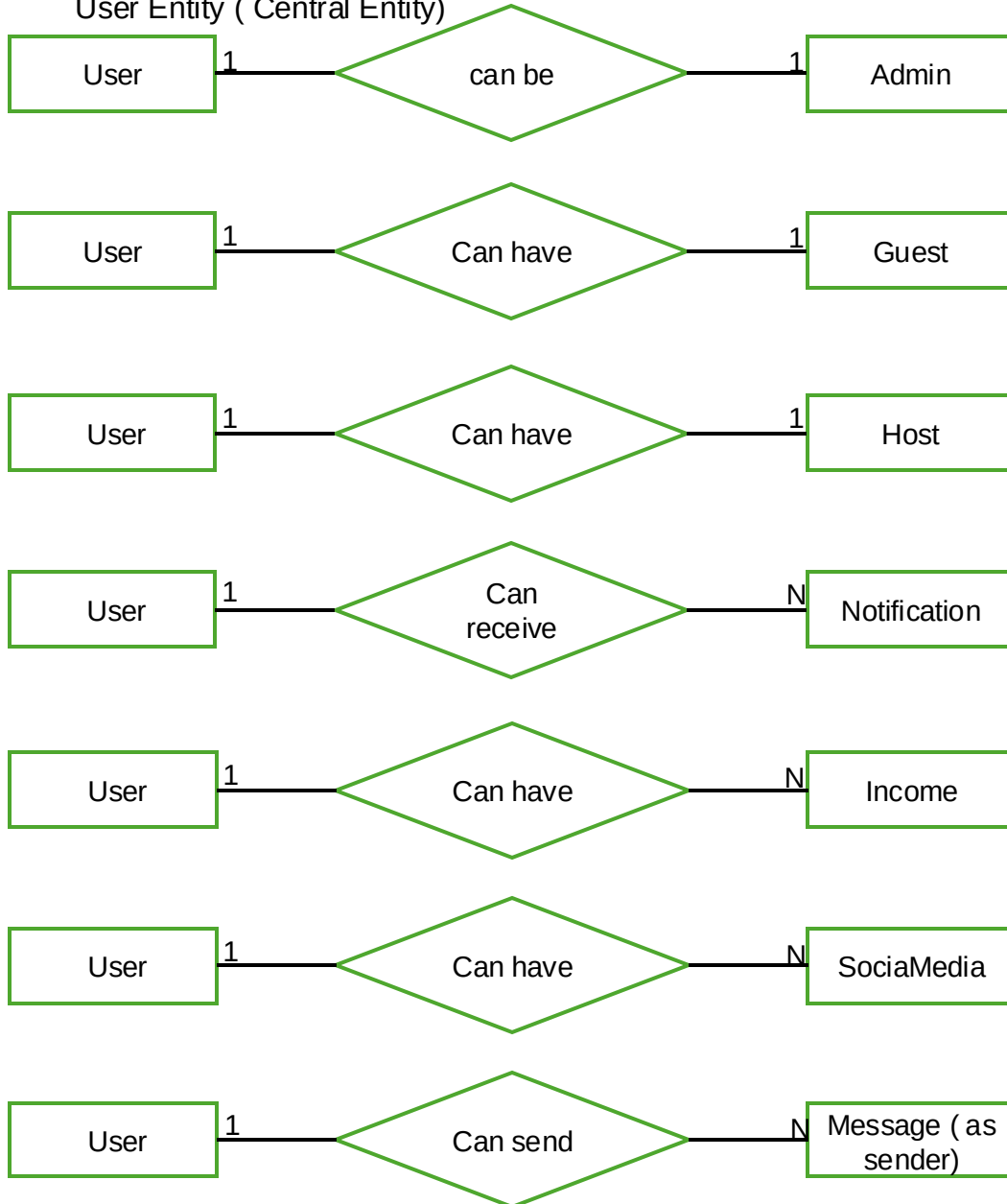
Table name	Column	Data type/ Max length
Admin	id(PK)	int auto_increment
	name	varchar(255) not null
	email	varchar(255) unique not null
	user_id(FK)	int not null
Notification	id(PK)	int auto_increment
	user_id(FK)	int not null
	message	text not null
	timestamp	datetime not null
Income	id(PK)	int auto_increment
	user_id(FK)	int not null
	source	varchar(255) not null
	amount	decimal(10, 2) not null
	income_date	date not null
Language	id(PK)	int auto_increment
	name	varchar(255) not null

Table name	Column	Data type/ Max length
UserLanguage	user_id(PK), (FK)	int auto_increment, int not null
	language_id(PK), (FK)	int auto_increment, int not null

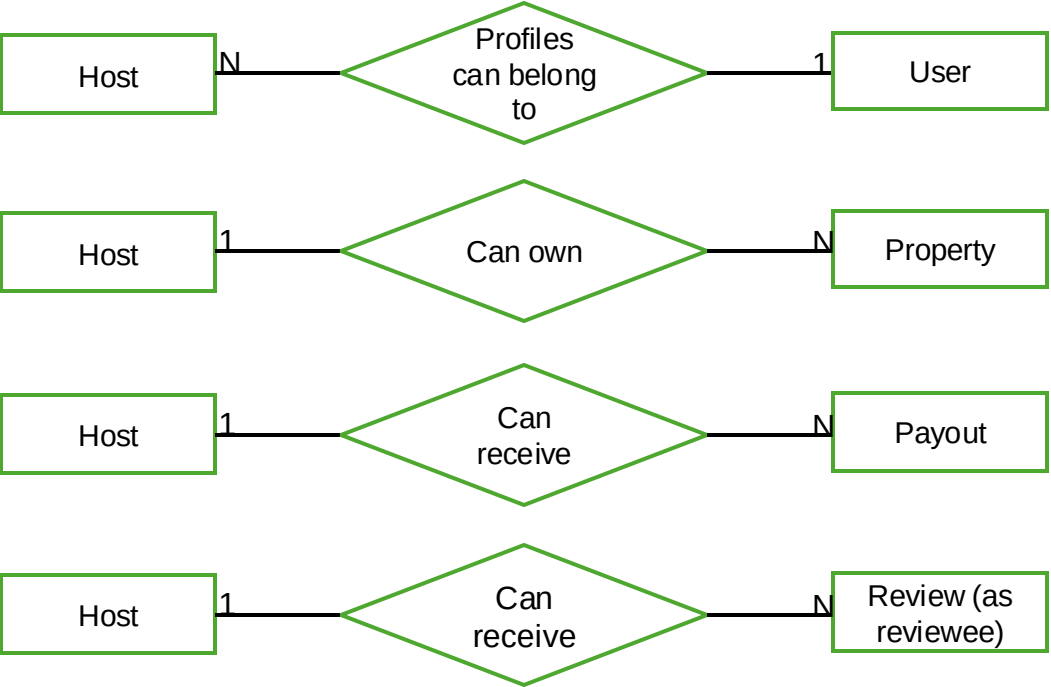
Cardinality:

Here is the cardinality of each table in the airbnb_company database schema:

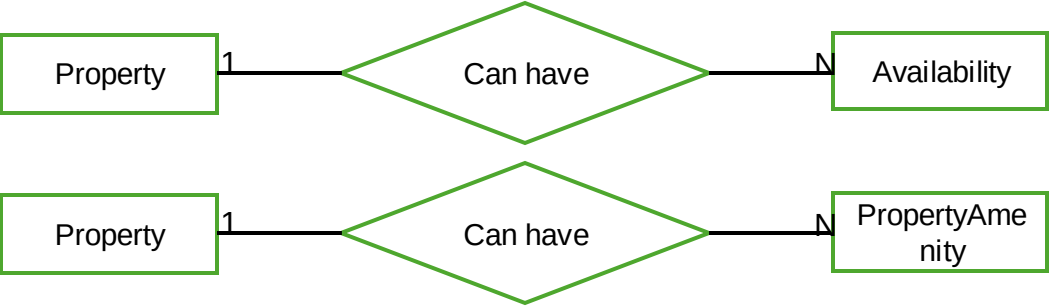
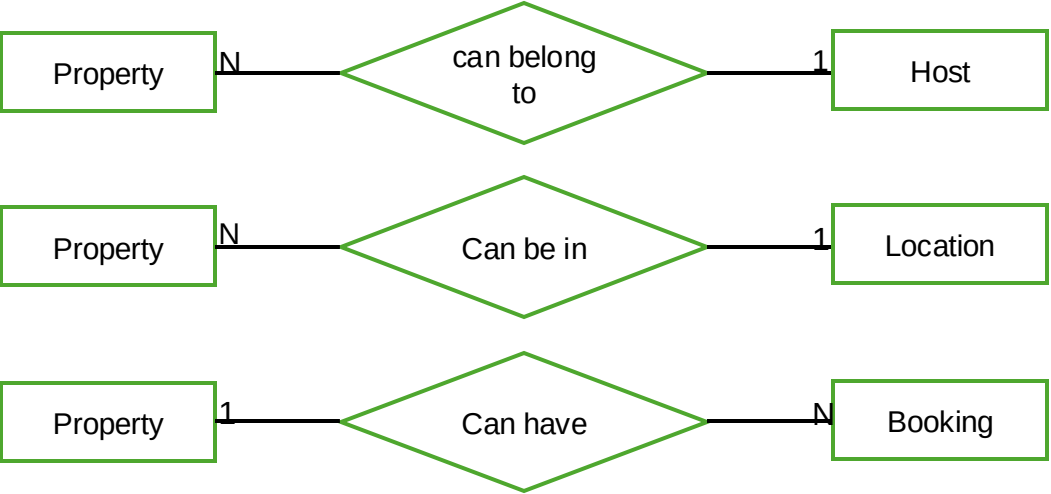
User Entity (Central Entity)



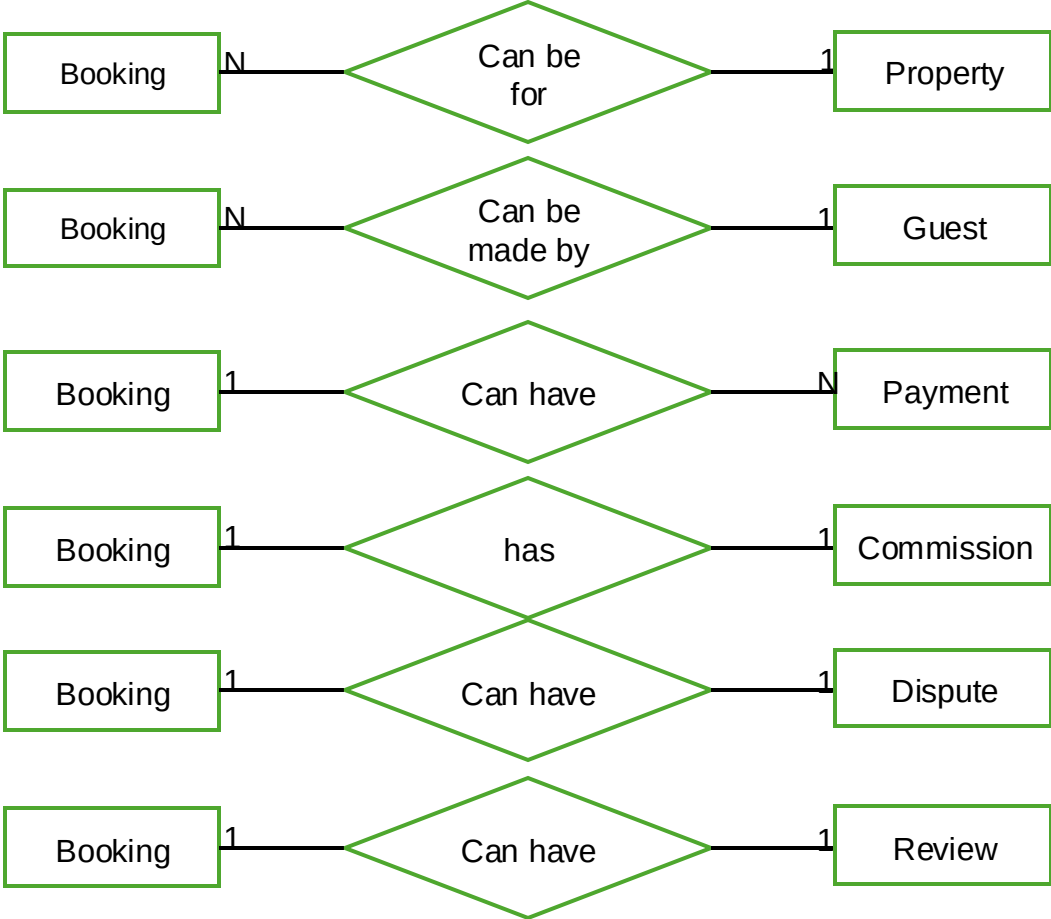
Host Entity:



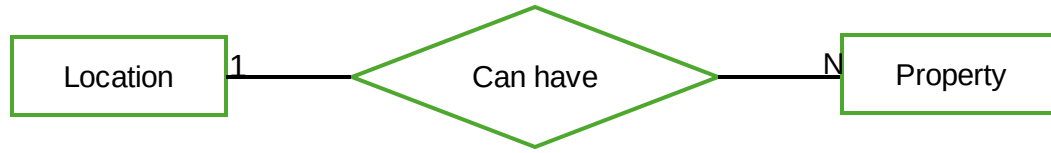
Property Entity:



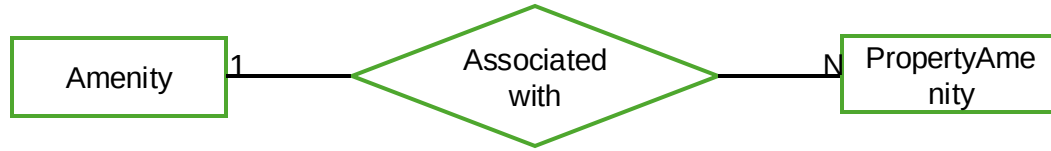
Booking Entity:



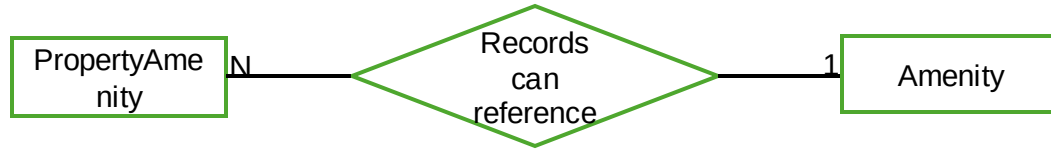
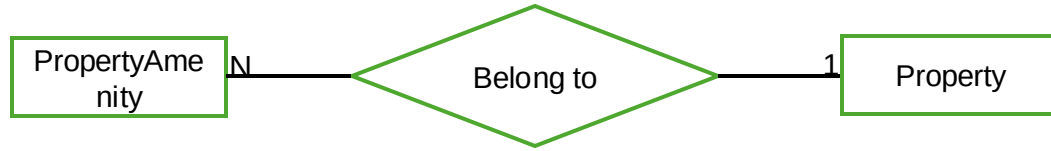
Location Entity:



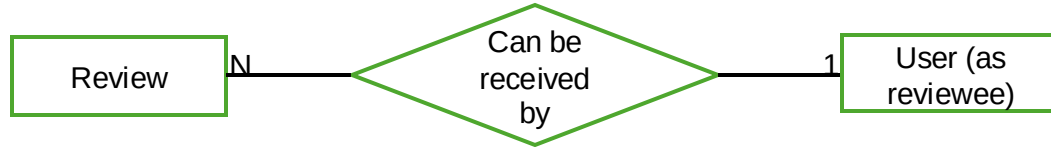
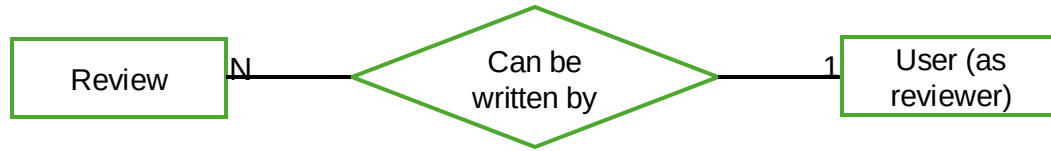
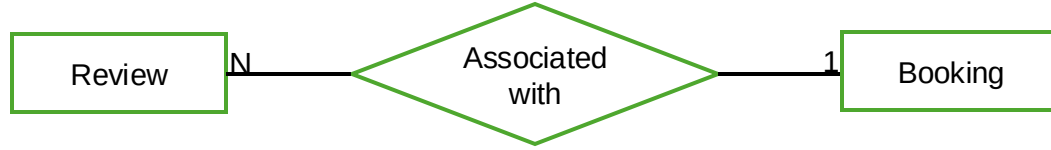
Amenity Entity:



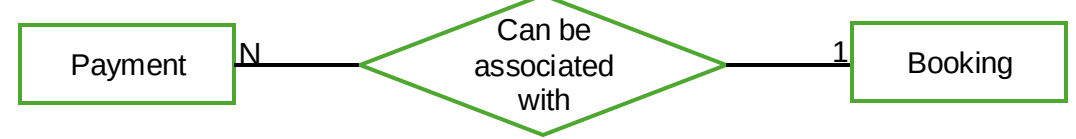
PropertyAmenity Entity: (Junction table)



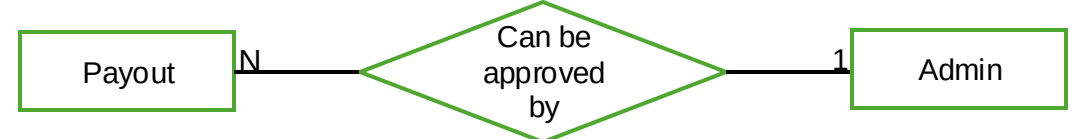
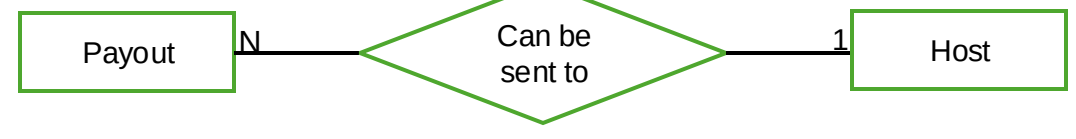
Review Entity:



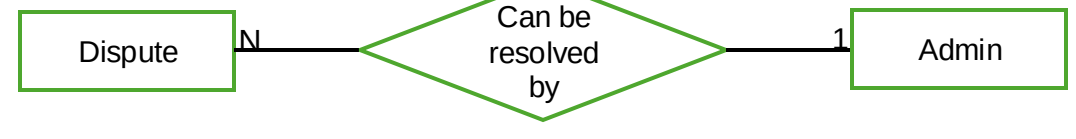
Payment Entity:



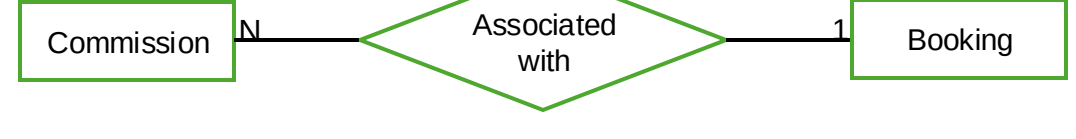
Payout Entity:



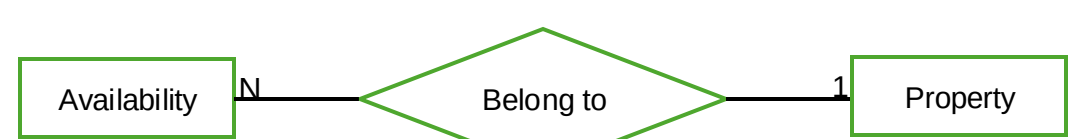
Dispute Entity:



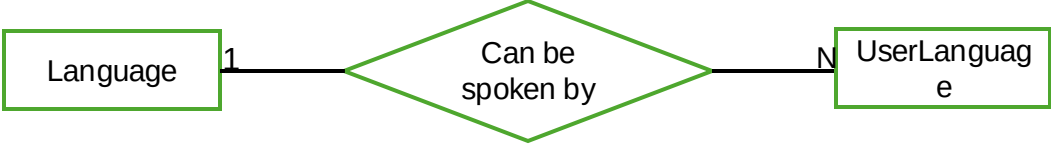
Commission Entity:



Availability Entity:



Language Entity:



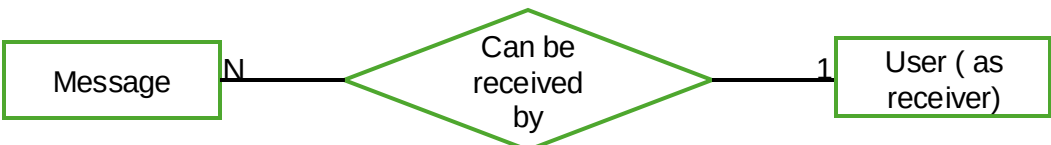
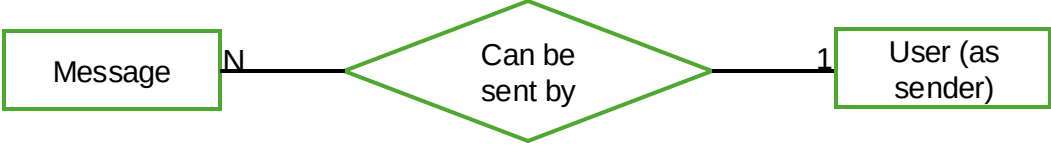
UserLanguage Entity: (Junction table)



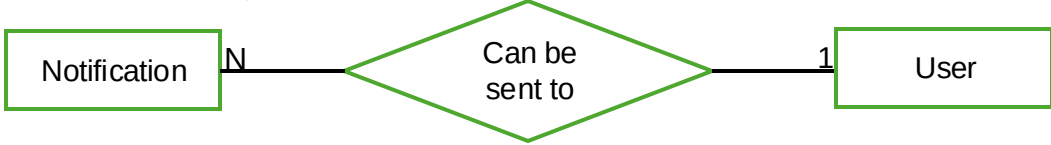
SocialMedia Entity:



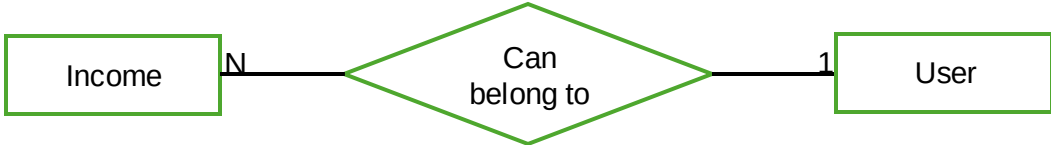
Message Entity:



Notification Entity:



Income Entity:



Information Schema:

	CATALOG_NAME	SCHEMA_NAME	DEFAULT_CHARACTER_SET_NAME	DEFAULT_COLLATION_NAME	SQL_PATH	DEFAULT_ENCRYPTION
	def	airbnb_company	utf8mb4	utf8mb4_0900_ai_ci	NULL	NO

	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	TABLE_TYPE	ENGINE	VERSION	ROW_FORMAT	TABLE_ROWS	AVG_ROW_LENGTH	DATA_LENGTH	MAX_DATA_LENGTH	INDEX_LENGTH	DATA_FREE	AUTO_INCREMENT
	def	airbnb_company	Admin	BASE TABLE	InnoDB	10	Dynamic	6	2730	16384	0	32768	0	6
	def	airbnb_company	Amenity	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	0	0	20
	def	airbnb_company	Availability	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	Booking	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	32768	0	20
	def	airbnb_company	Commission	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	Dispute	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	32768	0	20
	def	airbnb_company	Guest	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	Host	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	Income	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	Language	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	0	0	20
	def	airbnb_company	Location	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	0	0	20
	def	airbnb_company	Message	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	32768	0	20
	def	airbnb_company	Notification	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	Payment	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	Payout	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	32768	0	20
	def	airbnb_company	Property	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	32768	0	20
	def	airbnb_company	PropertyAme...	BASE TABLE	InnoDB	10	Dynamic	60	273	16384	0	16384	0	NULL
	def	airbnb_company	Review	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	49152	0	20
	def	airbnb_company	SocialMedia	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	User	BASE TABLE	InnoDB	10	Dynamic	20	819	16384	0	16384	0	20
	def	airbnb_company	UserLanguage	BASE TABLE	InnoDB	10	Dynamic	40	409	16384	0	16384	0	NULL

The total size of the database schema:

	Database_name	Total_Size_MB
	airbnb_company	0.73

List of tables in the database schema:

Tables_in_airbnb_company
Admin
Amenity
Availability
Booking
Commission
Dispute
Guest
Host
Income
Language
Location
Message
Notification
Payment
Payout
Property
PropertyAmenity
Review
SocialMedia
User
UserLanguage

The number of entries (row count) on each table in this schema

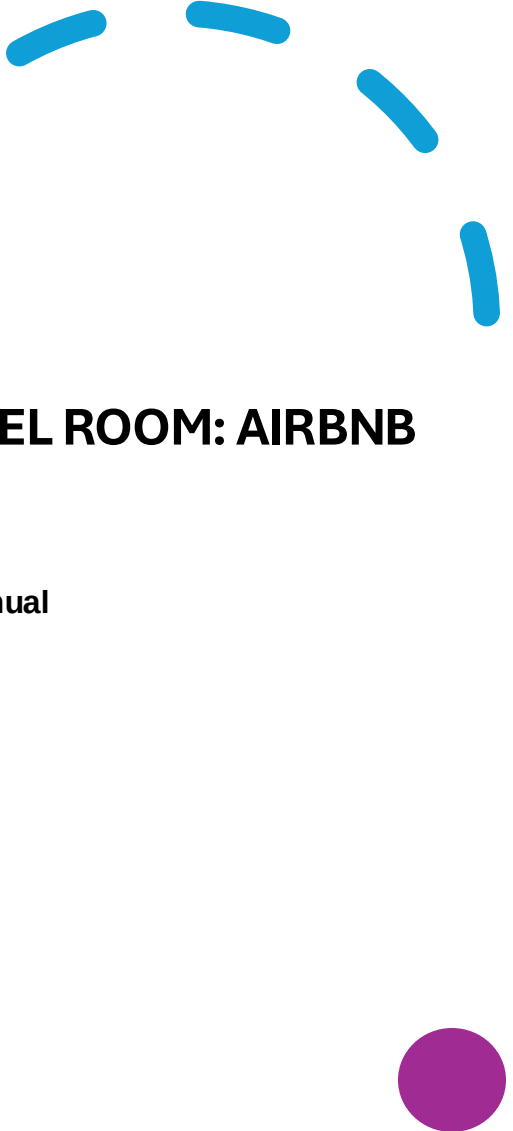
TABLE_NAME	TABLE_ROWS
Admin	6
Amenity	20
Availability	20
Booking	20
Commission	20
Dispute	20
Guest	20
Host	20
Income	20
Language	20
Location	20
Message	20
Notification	20
Payment	20
Payout	20
Property	20
PropertyAmenity	60
Review	20
SocialMedia	20
User	20
UserLanguage	40

Size of each table on schema
Descending order:

TABLE_NAME	table_size_MB
Review	0.06
Admin	0.05
Booking	0.05
Dispute	0.05
Message	0.05
Payout	0.05
Property	0.05
Availability	0.03
Commission	0.03
Guest	0.03
Host	0.03
Income	0.03
Notification	0.03
Payment	0.03
PropertyAme...	0.03
SocialMedia	0.03
User	0.03
UserLanguage	0.03
Amenity	0.02
Language	0.02
Location	0.02

List of foreign key constraints name and unique constraints name on the database schema:

CONSTRAINT_NAME	referencing_table	referencing_column	UNIQUE_CONSTRAINT_NAME	referenced_table	referenced_column
admin_ibfk_1	Admin	user_id	PRIMARY	User	id
admin_ibfk_1	Admin	user_id	PRIMARY	User	id
availability_ibfk_1	Availability	property_id	PRIMARY	Property	id
availability_ibfk_1	Availability	property_id	PRIMARY	Property	id
booking_ibfk_1	Booking	property_id	PRIMARY	Property	id
booking_ibfk_2	Booking	guest_id	PRIMARY	Guest	id
booking_ibfk_1	Booking	property_id	PRIMARY	Property	id
booking_ibfk_2	Booking	guest_id	PRIMARY	Guest	id
commission_ibfk_1	Commission	booking_id	PRIMARY	Booking	id
commission_ibfk_1	Commission	booking_id	PRIMARY	Booking	id
dispute_ibfk_1	Dispute	booking_id	PRIMARY	Booking	id
dispute_ibfk_2	Dispute	resolved_by	PRIMARY	Admin	id
dispute_ibfk_1	Dispute	booking_id	PRIMARY	Booking	id
dispute_ibfk_2	Dispute	resolved_by	PRIMARY	Admin	id
guest_ibfk_1	Guest	user_id	PRIMARY	User	id
guest_ibfk_1	Guest	user_id	PRIMARY	User	id
host_ibfk_1	Host	user_id	PRIMARY	User	id
host_ibfk_1	Host	user_id	PRIMARY	User	id
income_ibfk_1	Income	user_id	PRIMARY	User	id
income_ibfk_1	Income	user_id	PRIMARY	User	id
message_ibfk_1	Message	sender_id	PRIMARY	User	id
message_ibfk_2	Message	receiver_id	PRIMARY	User	id
message_ibfk_1	Message	sender_id	PRIMARY	User	id
message_ibfk_2	Message	receiver_id	PRIMARY	User	id
notification_ibfk_1	Notification	user_id	PRIMARY	User	id
notification_ibfk_1	Notification	user_id	PRIMARY	User	id
payment_ibfk_1	Payment	booking_id	PRIMARY	Booking	id
payment_ibfk_1	Payment	booking_id	PRIMARY	Booking	id
payout_ibfk_1	Payout	host_id	PRIMARY	Host	id
payout_ibfk_2	Payout	approved_by	PRIMARY	Admin	id
payout_ibfk_1	Payout	host_id	PRIMARY	Host	id
payout_ibfk_2	Payout	approved_by	PRIMARY	Admin	id
property_ibfk_1	Property	host_id	PRIMARY	Host	id
property_ibfk_2	Property	location_id	PRIMARY	Location	id
property_ibfk_1	Property	host_id	PRIMARY	Host	id
property_ibfk_2	Property	location_id	PRIMARY	Location	id
propertyamenity_ibfk_1	PropertyAmenity	property_id	PRIMARY	Property	id
propertyamenity_ibfk_2	PropertyAmenity	amenity_id	PRIMARY	Amenity	id
propertyamenity_ibfk_1	PropertyAmenity	property_id	PRIMARY	Property	id
propertyamenity_ibfk_2	PropertyAmenity	amenity_id	PRIMARY	Amenity	id
review_ibfk_1	Review	booking_id	PRIMARY	Booking	id
review_ibfk_2	Review	reviewer_id	PRIMARY	User	id
review_ibfk_3	Review	reviewee_id	PRIMARY	User	id
review_ibfk_1	Review	booking_id	PRIMARY	Booking	id
review_ibfk_2	Review	reviewer_id	PRIMARY	User	id
review_ibfk_3	Review	reviewee_id	PRIMARY	User	id
socialmedia_ibfk_1	SocialMedia	user_id	PRIMARY	User	id
socialmedia_ibfk_1	SocialMedia	user_id	PRIMARY	User	id
userlanguage_ibfk_1	UserLanguage	user_id	PRIMARY	User	id
userlanguage_ibfk_2	UserLanguage	language_id	PRIMARY	Language	id
userlanguage_ibfk_1	UserLanguage	user_id	PRIMARY	User	id
userlanguage_ibfk_2	UserLanguage	language_id	PRIMARY	Language	id



A USE-CASE OF BOOKING A HOTEL ROOM: AIRBNB

Database Installation Manual

A USE-CASE OF BOOKING A HOTEL ROOM: AIRBNB

This guidelines and installation manual provides the comprehensive database schema framework based on Airbnb system. It gives details information about the database schema along with instructions for installing and configuring the database system.

Database Schema Overview:

The airbnb_company database has in total 21 tables, they are:

Admin, amenity, availability, booking, commission, dispute, guest, host, income, language, location, message, notification, payment, payout, property, propertyamenity, review, socialmedia, user, userlanguage.

The database is built using MYSQL and consists of 21 interconnected tables designed to handle user roles (Host, Guest, Admin) , property, bookings, payments, reviews, and other functionalities of the platform.

The database facilities the user management having all the profile information for host and guests. Host can create and manage property listing, description of property where guest can easily search the property location, price and availability. The database can handle the booking made by guests. Host and guests should be able to communicate through the platform having the details of property and communication support. Both hosts and guests can rate, review and send or receive message each other after stayed on properties.

Details description of payment and payout should be recorded in database. The system can track the income generated through the bookings, providing the details transparent of revenue generation for hosts and the platform. User can handle the complain create on the platform or from the guest. The database can handle notifications send/receive to/by user.

Prerequisites:

MYSQL Server, MYSQL Workbench, and MYSQL Shell

Download and install MYSQL: download and install MySQL server, MySQL Workbench, and MySQL shell from the official website: [MYSQL Downloads](https://dev.mysql.com/downloads/).

Installation:

1. Open the MYSQL Workbench and set up a new connection.
2. Connect to the server you created in the previous step from MYSQL Workbench.
3. On the display screen, click on the server menu at the top and click on Data Import.
4. Select 'Import from Self-Contained File" option.
5. Next, select the save file from your device, and click on the small button with the two dots on the far right to browse to SQL file.
6. Browse to the location where is the 'airbnb_company' SQL file , select it, and click open.
7. Create a new schema at the 'Default Target Schema '
8. Please be sure 'Dump structure and Data' is selected.
9. Click on 'start import' button on the bottom right of screen.
10. You will see on screen ' Import Completed'.
11. Go to the 'schemas' tab and refresh it.
12. You will see the newly created schema and all the created tables.
13. If you want to make the test case of schema, you can execute the SQL file name 'test_case' after opening it from the 'Open a SQL script file in a new query tab.
14. You can also execute statements in the SQL file name 'metadata ' to access all the information schema of database.

Conclusion:

This manual provides the essential steps to install and configure the Airbnb database system. The database schema provides the overview details of tables structure and functionalities. The complete SQL script and documentation are included in the accompanying ZIP file.